مستشفى الملك فهد
King Fahad Hospital
تجمع الباحة الصحي

NURSING DEPARTMENT

Policy Name : Examination of Newborn Detailed Guideline

Policy Number : ESINU – NICU 6/3 (8)

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14. REVIEW SUMMARY

1. **Purpose**
    a. To have a clear guidance in ensuring the prevention and management of intravenous infiltration, Phlebitis and Extravasation is standardized in the pediatric intensive care unit.
    b. All staff act in accordance with this policy to prevent the development of intravenous Infiltration, Phlebitis and Extravasation.
2. **Scope**
    a. **First Step**
        i. Assess patient for any presence or risk for developing infiltration, Phlebitis and Extravasation.
    b. **Last Step**
        i. Document the assessment, preventive measures and management of infiltration, Phlebitis and Extravasation.
3. **Definition**
    a. **Phlebitis** is defined as an inflammation of the vein, is among the most common complications of IV therapy. Possible causes of phlebitis may be chemical (caused by the substance being infused), mechanical (movement of the catheter while inside the vein) or bacterial (either intrinsic or extrinsic).
    b. **Infiltration** is the leaking of IV fluid into the tissue surrounding the vein. This occurs when IV fluids continue to be delivered even though the tip of the catheter is no longer in the vessel or is blocked. Although most infiltrations do not cause tissue damage, the potential for serious harm does exist. When a large amount of IV solution enters the tissues, nerve compression injuries from compartment syndrome can result.
    c. **Extravasation** is an infiltration involving the leakage of vesicant fluids into the tissues. Vesicants are substances known to be more caustic than regular lV fluids, and are capable of causing physical harm unless they are kept strictly in the vascular space. These chemical or drug substances can cause blistering and burning in the tissue Vesicants include Calcium Chloride, Potassium Chloride, Sodium Chloride, Calcium Gluconate, Dobutamine, Diazepam, Dopamine, Norepinephrine (Levophed), Phenytoin (Dilantin), Promethazine (Phenergan), Propofol, Vasopressin, Radiologic contrast agents etc.
4. **Abbreviation:**
    a. IV - Intravenous
5. **Policy Statements**
    a. The infusion site must be inspected every 30-60 minutes and documented when in use and if extravasation or infiltration is suspected. Observe for any erythema or edema.
    b. The port needle entry site should be observed before administering vesicants ot irritant solutions.

c. The suggested maximum pressure alarm setting for an infusion pump is 15-25mmHg for vesicant drugs. The pressure of the infusion pump must be monitored.
d. The pressure alarm limit must be set when a vesicant infusion is commenced and rechecked at the beginning of each shift, if still running.
e. Record pump pressures and site monitoring on the fluid balance charts as per hospital policy.
f. A rise in pressure must be investigated.
g. The pressure reading should not be the sole indicator for an extravasation.
h. Bandages are not recommended for use when administering bolus vesicant therapy and should be used with caution for infusions. Never cover the insertion site as this compromises effective monitoring
i. Patient and family queries about pain, discomfort or swelling must be investigated; they should also have been informed of signs and symptoms

6. **Procedure and Rationale:**
   a. Materials/Equipment
   Below is an example of a typical extravasation kit, which included:
      i. Cold pack or hot pack
      ii. Antidotes according to local procedures
      iii. 2 mL syringes
      iv. Needles
      v. Skin disinfectant as per local guidelines (e.g., alcohol swabs)
      vi. Indelible pen for marking the affected area
      vii. Documentation forms

SN	PROCEDURE	RATIONALE
Preparatory Phase		
1.	Identify the patient by using two identifiers (Name and medical record number - first identification)	Correct identification of patient as per IPSG goal 1.
2.	**ASSESSMENT** Assess the factors known to increase the risk of IV site complications: **2.1. Patient Factors** a. Small blood vessels b. Fragile veins (e.g. cancer patients) c. Hard, sclerosed veins d. Mobile veins e. Impaired circulation f. Obstructed vena cava (elevated venous pressure can cause leakage)	It helps in early identification of factors lead to IV site complications and prevent the deterioration.

	g. Pre-existing conditions (diabetes, peripheral circulatory conditions like Raynaud's syndrome, radiation damage) h. Obesity **2.2.Cannulation and Infusion Procedure:** a. Untrained or inexperienced staff b. Multiple attempts at cannulation c. Unfavorable cannulation site (e.g., back of hand vs. forearm, close to bone) d. Bolus injection e. High flow pressure **2.3.Medication Factors:** a. Vesicant potential b. Volume of drug/fluid infiltrated c. Concentration of vesicant drug/fluid d. Osmolarity of drug/fluid e. Vasoconstrictive potential f. Cytotoxicity **2.4.Peripheral Cannula Related:** a. Metal/steel needles (butterfly) b. Large gauge cannula relative to vein size. c. Inadequately secured cannula d. Undesirable cannula site location (e.g. antecubital fossa, dorsum of hand or wrist rather than forearm, areas of joint flexion and use of dominant hand). e. Clot formation above cannula site	
PREVENTION OF INFILTRATION, PHLEBITIS AND EXTRAVASATION		
3.	**3.1.Skilled Cannulation**	
	a. All nurses should be proficient in IV cannulation	lf cannula inserted without difficulty the risk of fluid leakage from the vein is reduced.
	b. Should have only 2 attempts at cannulation	Avoiding multiple venipuncture decreases the chance of IV fluid leaking from the veins.
	c. Avoid small fragile veins	Veins that is small and fragile at greater risk of perforation.

	d. Avoid cannulating areas of flexion; cubital fossa or wrist	Areas of flexion have greater movement which can potentially cause the cannula to dislodge.
	e. Multiple attempts at cannulation; each attempt should be higher up the arm than the previous attempt.	When cannulation has been attempted there is a potential that vein has been pierced and for IV fluid to leak into the tissue through the breach.
	f. Use the smallest gauge and shortest length cannula.	Helps improve blood flow around the cannula and to move IV fluid into the blood stream.
	g. Use a freshly sited cannula.	Ensures cannulas are at their greatest patency.
	h. Always have full visibility of IV cannula site and ensure cannula is adequately secured in place.	Complete visibility allows quicker identification of potential extravasation.
	i. Administer drugs with vesicant potential first.	Ensures these drugs are given when iv cannula is at its greatest patency.
	j. Stay with/in vicinity of the patient whilst the vesicant is being administered	lf extravasation occurs appropriate action can be taken as quickly as possible, potentially decreasing the side effects.
	a. Verify patency of iv cannula prior to, during and after administration of the cytotoxic ie assess for blood return	Ensure the cannula is Functioning. lf there is no blood return it may indicate incorrect positioning of cannula.
	3.2.Early Identification of Symptoms. Any or all of the following are signs of extravasation: a. Bleb formation b. Indurations, erythema, venous discoloration or swelling c. Observed at the site. d. The patient complains of burning, stinging, pain or any acute change at the injection site. e. Alteration to the rate of flow or increased resistance to the Administration of the cytotoxic medication.	Early identification of signs and symptoms of IV site complications help to manage it at the initial stage effectively.

	f. No blood return is obtained g. Leakage of fluid from around injection site.	
	3.3.Monitoring	
	a. Check for swelling, inflammation, redness and pain around cannula site during administration of IV drugs.	To get baseline data
	b. Check blood return from the cannula when vesicants are administered.	To ensure patency of peripheral line
	c. Question the patient about any possible symptoms (i.e., heat, pain and swelling during administration).	Subjective experience should be collected and evaluated
	d. Do not allow patients receiving intravenous infusions of vesicant drugs to leave clinical area Considerations for vesicants.	Frequent monitoring of IV site is indeed
	e. Whenever possible, always give vesicant drugs into a recently inserted cannula.	It decreases chance for IV site complications
	f. Patients receiving repeated doses of potentially harmful drugs peripherally should have the cannula resited at regular intervals.	It decreases chance for IV site complications
	g. A central vein could be considered if veins are very difficult to access.	Administration of vesicants minimize the risk of extravasations
Performance Phase		
1.	**MANAGEMENT OF PHLEBITIS INFILTRATION AND EXTRAVASATION** **1.1.Phlebitis:** a. An affected limb should be elevated b. Anti-inflammatory analgesics can be administered as per order	To minimize inflammation To treat both inflammation and the associated with phlebitis.
	1.2.Infiltration and Extravasation: At the first indication of extravasation or if an extravasation is suspected: a. Stop the infusion/injection immediately. b. b. Wear PPE when dealing with a cytotoxic extravasation.	To prevent deterioration in infiltration To take infection control precautions To reduce anxiety and get cooperation To aspirate the fluid from the site

	c. Explain incident to patient and reassure. d. DO NOT remove the cannula. e. Disconnect infusion. f. Connect 10ml syringe and attempt to aspirate as much blood/fluid as possible from the cannula. g. Do not exert pressure on the extravasated area. h. Elevate limb and immobilize Elevation of the affected limb i. Contact attending physician and treat as per orders. j. lf there is no antidote, remove the cannula. k. Mark affected area. l. Apply a cold pack or warm compress to the affected area for 20 minutes 4 times daily for 1-2 days as per doctor's advice and characteristics of drugs. m. Assess site for pain, erythema, in duration and necrosis. n. Administer Antihistamines and analgesics as per doctor's order. o. Administer Corticosteroids as per doctor/s order.	Disconnect and keep infusion for infusing through another line To remove extravasated fluid as much as possible It damages tissue It may aid in reabsorption of the infiltrate or extravasated vesicant by decreasing capillary hydrostatic pressure. Review and initiate the appropriate treatment for the cytotoxic medication administered at the time of incident. Local cooling (ice packs) aids in vasoconstriction, thus theoretically limiting drug dispersion. Local warming therapy (dry heat) enhances vasodilation, thus enhancing dispersion of the vesicant agent and decreasing drug accumulation in the local tissue. It provides baseline data. It may be required for relief of pain and other symptoms. It can be given to treat inflammation.
Follow-up Phase		
1.	Monitor site for pain, erythema, indurations and necrosis at intervals recommended by the doctor and treat accordingly.	Note that extravasations reactions may not manifest until some hours following the infusion.
2.	Complete incident form, and document event details in the patient medical record.	

Infiltration Scale

Use these classifications when documenting infiltration

Grade 0	No symptoms
Grade 1	1. Skin blanched 2. Edema less than 1 inch (2.5 cm) in any direction 3. Cool to touch 4. With or without pain
Grade 2	1. Skin blanched 2. Edema 1to 6 inches (2.5 to 15 cm) in any direction 3. Cool to touch 4. With or without pain
Grade 3	1. Skin blanched, translucent 2. Gross edema more than 5 inches in any direction 3. Cool to touch 4. Mild to moderate pain 5. Possible numbness
Grade 4	1. Skin blanched, translucent, tight, leaking, discolored, bruised, swollen

Phlebitis Scale

0	No Symptoms
1	Erythema at access site with or without pain
2	Pain at access site with Erythema and/or edema
3	Pain at access site with Erythema and/or edema, streak formation, palpable venous cord
4	Pain at access site with Erythema and/or edema, streak formation, palpable venous cord greater than 1-inch in length; purulent drainage

7. **Responsibilities**

a. **Physician** is responsible in ensuring that drugs and dilution is properly written in physician's order.
b. **Clinical Instructor** is responsible to conduct knowledge based competency lectures checks off for the newly joined staff and supervises the actual performance of the unit.
c. **Nursing supervisor** is responsible in collecting and submitting the report on patients with IV site complications and responsible in collecting the OVR and submit to the Quality department within 24 hours.
d. **Head Nurse/ Charge Nurse** are responsible in ensuring that IV lines are patent. Encourage the bedside staff to monitor and identify the early signs of IV site complications. Follow up of OVR is one of the other responsibilities.
e. **Staff Nurse** is responsible in maintaining IV site in secured way and monitoring any signs of infiltration for preventing IV site for any signs of complications. Report to Charge nurse and/or
f. **Head nurse** any complications developed and, manage it properly and make OVR

8. **Documentation**
   a. Document the following in nurse's progress note and OVR Report:
      i. IV line location, size, date of insertion
      ii. Discoloration
      iii. Measurement (length, width and depth) of infiltration
      iv. Condition of the surrounding skin
      v. Care given for iv site complications
      vi. Drainage
      vii. Patient's tolerance
9. **Measurements**
   a. 100% of the nurses should know the protocol of blood transfusion.
10. **References**
   a. Ministry of Health, Policy and Procedure Manual, KSA
   b. Infusion Nursing Standards of Practice 2011
   c. Hadaway, L. (2007) Infiltration and extravasation Am J Nurs 107 (8):64-72.
   d. http://tangentmedical.com/iv-complications/
   e. Sauerland C, Engelking C, Wickham R, Corbi D (2006) Vesicant extravasation part I: Mechanisms, pathogenesis, and nursing care to reduce risk. Oncol Nurs Forum 33 (6): 1134-41.
   f. Source: From the Infusion Nursing Standards of Practice 2011
11. **Cross References**
    N/A
12. **Attachments**
   a. Narcotic form
   b. Expiry dates of the drugs check list.
13. **Distribution List**
   a. Nursing administration

b. Quality department

14. **Review Summary**
    a. Revision of Purpose statement.
    b. Reformat of policy statement.